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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Session	**2025-26 (Even)**	**Course Name**	**PE-III: Lab -Reinforcement Learning**
Semester	**6th**	**Course Code**	**23ADS1644**
Roll No	**42**	**Name of Student**	**Krishna Mishra**

Practical Number	**03**
Course Outcome	CO1: Apply fundamental reinforcement learning concepts to model and solve decision-making problems. CO2: Implement basic reinforcement learning algorithms to train agents in a given environment. CO3: Analyze agent behavior
Aim	To implement linear regression to predict continuous values and analyze the relationship between independent and dependent variables.
Problem Definition	The objective of this experiment is to implement a Linear Regression model to predict continuous values using a real-world dataset. Specifically, the task is to analyze and model the relationship between the independent variable **total rooms** and the dependent variable **median house value** from the California housing dataset. The goal is to train a predictive model, evaluate its performance using appropriate metrics, and visualize the relationship between variables using plots.
Theory (100 words)	Linear Regression is a supervised machine learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It assumes a linear relationship of the form y=mx+cy = mx + cy=mx+c, where mmm is the slope and ccc is the intercept. The algorithm minimizes the error between predicted and actual values using techniques like Least Squares. Performance is evaluated using metrics such as Mean Squared Error (MSE) and R-squared (R^2). MSE measures prediction error, while R^2 indicates how well the model explains variance in the data. Linear Regression is widely used due to its simplicity, interpretability, and effectiveness in predicting continuous numerical outcomes.

Procedure and Execution (100 Words)	Algorithm: 1. Import required libraries (NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn). 2. Load the dataset using Pandas. 3. Select independent variable (total_rooms) and dependent variable (median_house_value). 4. Visualize the data using a scatter plot. 5. Split the dataset into training and testing sets (80% training, 20% testing). 6. Initialize the Linear Regression model. 7. Train the model using the training data. 8. Predict the output using the test data. 9. Evaluate the model using MSE and R^2 score. 10. Display model parameters (intercept and coefficient). 11. Plot the regression line along with actual data points.

Code:

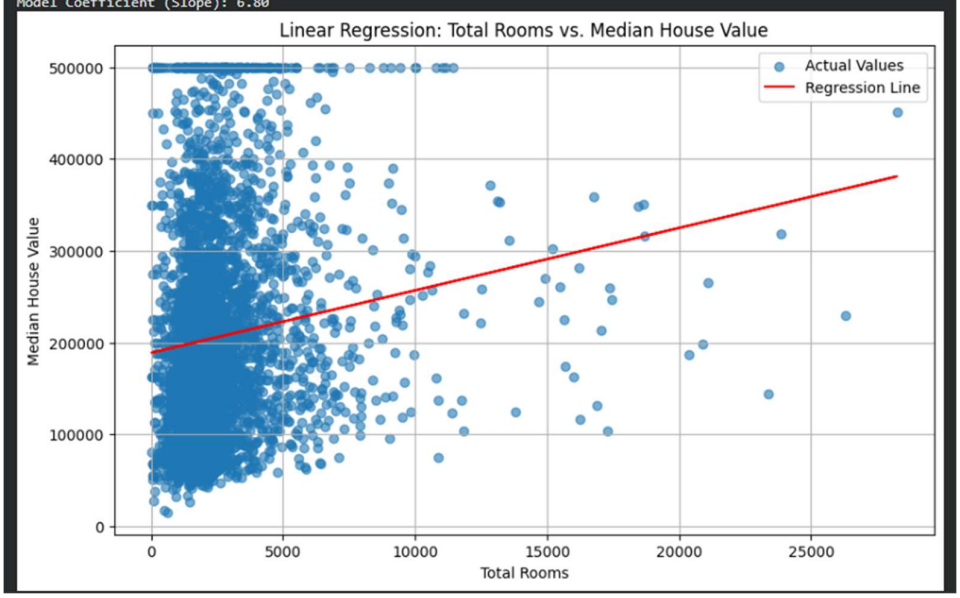
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_csv('/content/sample_data/california_housing_train.csv')
print("Dataset loaded successfully. First 5 rows:")
display(df.head())

X = df['total_rooms'].values.reshape(-1, 1)
y = df['median_house_value']
print(f"Shape of X: {X.shape}, Shape of y: {y.shape}")

plt.figure(figsize=(10, 6))
sns.scatterplot(x='total_rooms', y='median_house_value', data=df)
plt.title('Scatter Plot of Total Rooms vs. Median House Value')
plt.xlabel('Total Rooms')
plt.ylabel('Median House Value')
plt.grid(True)
```

```
plt.show()

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
print(f"Training data shape: X_train={X_train.shape},
y_train={y_train.shape}")
print(f"Testing data shape: X_test={X_test.shape}, y_test={y_test.shape}")

model = LinearRegression()
model.fit(X_train, y_train)
print("Model training complete.")

y_pred = model.predict(X_test)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"R-squared (R2) Score: {r2:.2f}")
print(f"Model Intercept: {model.intercept_:.2f}")
print(f"Model Coefficient (Slope): {model.coef_[0]:.2f}")

plt.figure(figsize=(10, 6))
plt.scatter(X_test, y_test, label='Actual Values', alpha=0.6)
plt.plot(X_test, y_pred, color='red', label='Regression Line')
plt.title('Linear Regression: Total Rooms vs. Median House Value')
plt.xlabel('Total Rooms')
plt.ylabel('Median House Value')
plt.legend()
plt.grid(True)
plt.show()
```

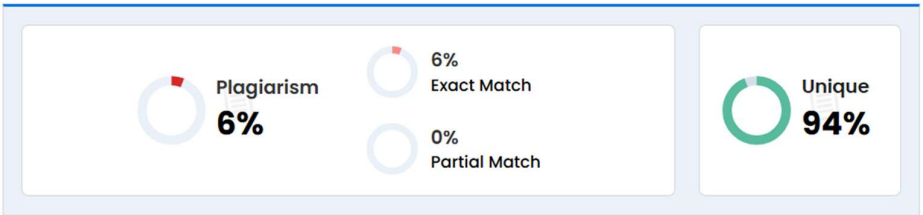
Output:

1. **Linear Regression Graph**

```
Training data shape: X_train=(13600, 1), y_train=(13600,)
Testing data shape: X_test=(3400, 1), y_test=(3400,)
Model training complete.
Mean Squared Error (MSE): 13504570751.71
R-squared (R2) Score: 0.02
Model Intercept: 188752.10
Model Coefficient (Slope): 6.80
```



Output Analysis	• The scatter plot shows a positive correlation between total rooms and median house value, though not perfectly linear. • The regression line represents the predicted relationship between variables. • The **Mean Squared Error (MSE)** indicates the average squared difference between actual and predicted values; lower values indicate better performance. • The **R^2 score** reflects how well the model explains the variance; a value closer to 1 indicates a better fit. • The coefficient (slope) shows how much the house value increases with an increase in total rooms. • Since only one feature is used, the model may not capture all influencing factors, leading to moderate accuracy.
Link of student Github profile where lab	https://github.com/krish-na45/RL_Practicals

assignment has been uploaded	
Conclusion	The Linear Regression model successfully predicts median house values based on total rooms and demonstrates the relationship between variables. While the model provides a basic understanding and reasonable predictions, its performance is limited due to the use of a single feature. Incorporating more relevant features such as location, income, and population could significantly improve accuracy. Overall, this experiment highlights the usefulness and simplicity of Linear Regression in predictive analysis.
Plag Report (Similarity index < 12%)	
Date	28/03/2026